Natural Computing 1

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Abstract

We investigate 1-dimensional binary-state cellular automata (CA) and their behaviours based on transition rules. We categorize these CAs according to Wolfram's classification system, which divides them into four distinct classes. We assess the nature of different CA rules, including Rule 30, Rule 136, Rule 110, and Rule 18. Additionally investigate the usage of CAs in modern research. Our study emphasizes the patterns, behaviours, and applications of CAs in both theoretical and practical contexts.

1 Introduction

For assignment 1 of Natural Computing, we focused on a 1-dimensional, binary-state cellular automata. These simple CAs show a diverse range of behaviours based on the rules used for determining the state of a cell. This paper presents our implementation of a 1-dimensional, binary-state CA, detailing the tests performed and our findings in the context of Wolfram's classification system.

Background

Stephen Wolfram, studied cellular automata and introduced a classification system that categorizes them in four classes based on their behaviour [3]. These Wolfram classes offer a systematic way of understanding the evolution of CAs.

CAs are also used in modern research. One instance of this is *"A cellular automaton model for tumor dormancy: Emergence of a proliferative switch"* by Chen D. et al. [1]. For the research they used a CA to understand tumor dormancy, specifically investigating a sudden switch in a tumor when it starts dividing rapidly after being dormant.

Further, for those new to the concept of CAs, Weisstein [2] provides an overview of the fundamental principles and behaviours of CAs in an easy to understand format, accessible for all audiences.

2 Problem Description

To understand the report its important to understand what a cellular automaton is. A cellular automaton [2] is a simulation of a system on a grid of cells that depicts its evolution over time based on the states of the cells. In 1-dimensional binary-state cellular automata, the simulation is conducted on a linear array, where each cell in the array can have one of two possible states in our case this is 1 or 0.

2.1 Transition rules

In 1-dimensional binary automata, there are 8 possible transitions for a given cell. These transitions determine the state of a cell in the next generation based on its current state and the states of its neighbors. 010, for example, would mean that the left neighbour is 0 and the current cell is 1. below is a table with the example rule 30.

Current Pattern	111	110	101	100	011	010	001	000
Next State	0	0	0	1	1	1	1	0

Table 1: Transitions for Rule 30

The reason this transition set is called rule 30 is because the next state can also be seen as 00011110, which is 30 if it is converted from binary to decimal. Later in the paper figure 5 shows a visualization of rule 30 in our CA program, the pattern breaks since our world can only be of width 60.

2.2 Wolfram classes

Stephen Wolfram describes four different "wolfram classes". These classes are behaviours that can be observed in these automata.

Wolfram class 1 are cellular automata who end up in a stable state. Meaning that it leads to a configuration where there is no more change observed after each generation. This generally means that there is loss of information about the initial state.

Wolfram class 2 can be observed whenever there are periodical local structures. Cyclic behaviour is observed.

Wolfram class 3 are cellular automata where there is no predictability. Deterministic chaos ensues.

Wolfram class 4 are observed whenever there is structure, but not necessarily periodical. This lays in between cyclic systems and deterministic chaos.

To decide which transition rule-set belongs to which class is ambiguous. This is for the same reason as the halting problem. For certain automata, we do not know if the world will end up in a loop or repeatedly be unpredictable for example.

3 Results

For each graph, the horizontal axis represents the generation and the vertical axis represents the count of which is described in the caption. For each line we show the average of 50 different iterations with the same rule.

3.1 Each wolfram class in action

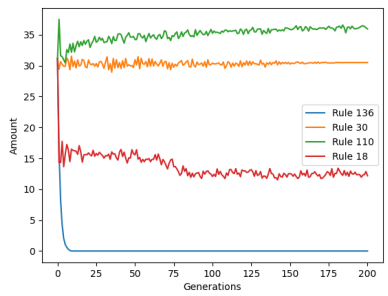

Figure 1: Total amount of 1s

In figure 1, it becomes clear that rule 136 completely dies out and is left with only zeroes, indicating that the world living by rule 136 is **Wolframm Class 1**. From the other rules we cannot draw a conclusion simply from this data.

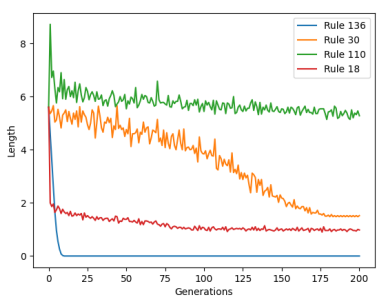

Figure 2: Longest consecutive string of 1s

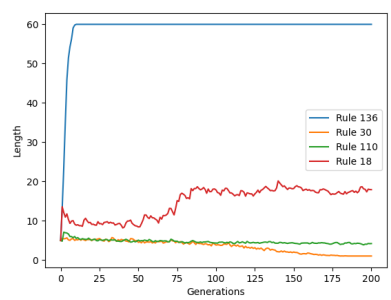

Figure 3: Longest consecutive string of 0s

In figure 2 and 3 we can see the longest string of 0s or 1s. The result of rule 136 is obvious, as the world dies out. However, we can also see that rule 30 slowly decreases. Rule 18 and 110 are relatively stable. We cannot draw many conclusions from these graphs, but it is possible to understand what is happening in the world later on, and these graphs will make sense by then.

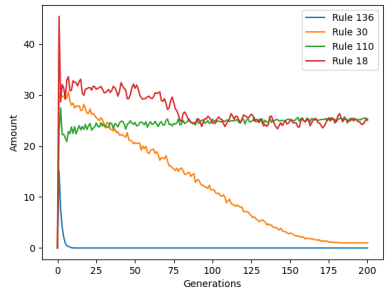

Figure 4: Amount of changes compared to last generation

In figure 4 we can see the amount of change. Interestingly, rule 30 dies out. In figure 5 we see a visual representation of what happens to rule 30, and now it becomes obvious as to what happens: eventually, the neighbours cancel each other out and we are left with consistent "lines", which cannot interact with each other, resulting in a dead world. This is a case of **Wolfram Class 2**. Rule 110 and 18 appear to constantly change each generation, but we are not observing whether it is random, or structured. Meaning, we cannot classify these yet.

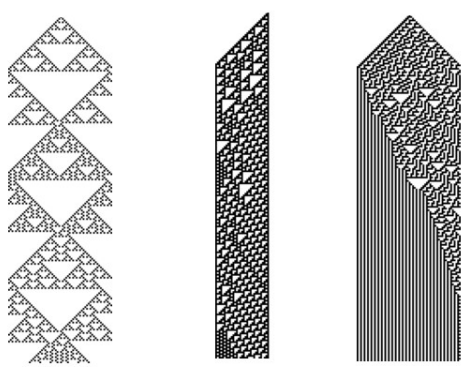

Figure 5: Rule 18, 110 and 30 visualized respectively

In figure 5 we visualized rule 18 and 110, and it immediately becomes clear that rule 18 is of **Wolfram class 4**, because there is a repeating structured never-ending pattern. Rule 110 is a bit more complicated, but if observed properly, there is no pattern. The triangles are of "random" size. Therefore, rule 110 can be said to be **Wolfram class 3**. As mentioned earlier, it is an ambiguous matter, because we cannot say that rule 110 will never repeat.

3.2 Langton Factor

Another interesting thing about 1-dimensional CAs is the Langton Factor. This determines the amount of living cells in the initial state. We will be testing with a factor of 1 gives the below graphs.

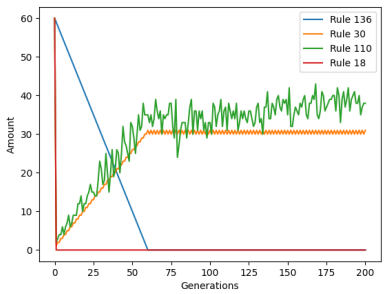

Figure 6: Total amount of 1s

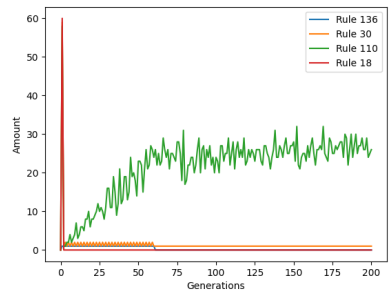

Figure 7: Total changes per generation

Comparing these graphs to the ones we obtained before will show that our original graphs are close to being Langton Factor 0.5. This is because we originally repeated the process 50 times with random initialized states. This will of course move towards the mean, which is 0.5. It is also interesting to see that all the rules stabilize quickly except for rule 110, this probably has to do with it being a wolfram class 4 rule.

4 Application of CA in practice

A modern implementation of a CA is used in *"A Cellular Automaton Model for Tumor Dormancy: Emergence of a Proliferative Switch"* by Chen et al. [1].

The researchers used a 2D CA model to study the growth of tumors. The cells could have multiple states: Dormant, actively dividing and dead. The cell transitions are based on multiple parameters, they can be classified in 2 different categories: Micro environmental parameters (based on the tumors environment/neighbouring cells) and Tumor growth parameters (based on parameters of the tumor/cell itself). based on these parameters Critical threshold values can be calculated, and these values will show if the cell will be dormant, killed or if it will continue dividing.

The field of study is theoretical oncology, specifically focusing on the behaviour of tumors during dormant and proliferative states. The context revolves around the longstanding issue of cancer dormancy and the phenomenon where tumors remain inactive for long periods before rapid growth.

The researchers chose for a CA because modeling allows for the exploration of various scenarios for potential dormancy mechanisms and their outcomes. The chosen CA model enables the simulation of interactions between the tumor and the surrounding micro environment. The CA model induces a "competition" between the tumor's

growth propensity and the micro environmental factors suppressing its growth.

The researchers were successful in their approach. The CA model replicated the dormancy period observed in tumor growth and demonstrated the switch behaviour where the tumor moves from a dormant state to rapid dividing state. By manipulating parameters like suppression rate, transformation rate, and micro environment rigidity, the researchers could determine their effects on tumor growth dynamics.

We think the research shows mechanisms leading to the "switch" behaviour comprehensively. Given the complexity of the way tumors work and interact, the use of a CA model is logical, as it allows for testing and showing what happens based on different scenarios and when changing parameters. The ability of the model to show such interactions tells us that it might bring more understanding to the way tumors work.

The current model is 2D for simplicity, we think an improvement would be to translate it to a 3D environment since a 2D environment might not be representative of a persons body, or the shape of organs in a human body. They also compare their results with theoretical data, it might benefit the research if it was compared to real-world data.

5 Conclusion

This study delved into the intricacies of 1-dimensional binary-state cellular automata, using Wolfram's classification system as a framework for understanding their varied behaviours. Our findings illustrate the diversity in CA behaviours, ranging from static patterns to deterministic chaos, based on their rule sets. Particularly, we observed that the behaviours of Rule 30, Rule 136, Rule 110, and Rule 18 align with Wolfram's Classes 1, 2, 3, and 4, respectively. Additionally, the real-world implications of CA models in understanding tumor behaviour were explored through the work of Chen et al. Their application underscored the potential of CAs in simulating complex systems and provided insights into possible mechanisms for tumor dormancy and proliferation. Future research could extend these findings by modeling CAs in three-dimensional environments or by correlating theoretical results with empirical data.

References

[1] Chen, D., Jiao, Y., Torquato, S.: A cellular automaton model for tumor dormancy: Emergence of a proliferative switch. PLoS ONE **9**(10), e109934 (2014). https://doi.org/10.1371/journal.pone.0109934, `https://doi.org/10.1371/journal.pone.0109934`, accessed: 18 October 2023

[2] Weisstein, E.W.: Cellular automaton `https://mathworld.wolfram.com/CellularAutomaton.html`, accessed: 18 October 2023

[3] Wolfram, S.: Statistical mechanics of cellular automata. Reviews of modern physics **55**(3), 601 (1983)